

Chapter 1

Title

Authorname

Abstract

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This chapter illustrates the factors that render Italian nouns and adjectives uninflectable. Three phonological factors – ending in a stressed vowel, in a consonant or in /i/ in the citation form – are causes of uninflectability for both nouns and adjectives; while ending in a stressed vowel results in exceptionless uninflectability at all times, nouns and adjectives ending in a consonant were phonologically adapted by providing them with a final vowel, and subsequently integrated in an inflecting class, in older stages of the Italian language, but at present this does not happen anymore; a few nouns and adjectives ending in /i/ have occasionally developed an inflecting variant through backformation, when the form ending in /i/ has been interpreted as an inflected masculine plural form. Another factor that produces uninflectability of nouns is a mismatch between a noun's final vowel and its gender: neological masculine nouns in /a/ and feminine nouns in /o/, whose final vowels are typical of nouns bearing the opposite gender's value, are treated as uninflectable in contemporary Italian. Even recently borrowed or recently coined nouns ending in the vowels most typical for their gender can be uninflectable; this is more common for nouns denoting inanimate entities, and for nouns bearing foreign characteristics in their spelling. There are also cases of constructional uninflectability, two of which are briefly discussed. The chapter also illustrates the inflectional classes of Italian nouns and adjectives and provides quantitative data about types and tokens in each inflectional class from several available databases.

Keywords: Italian, Nouns, Adjectives, Uninflectability, Diachrony

1 Introduction

Spencer2020 introduced a useful terminological distinction between *uninflecting* and *uninflectable* lexemes. He calls *uninflecting* lexemes and classes of lexemes that “are not associated with any inflectional paradigm”, like English prepositions, and *uninflectable* “members of an otherwise inflecting class” that “unexpectedly fail to inflect”, like the Russian noun KENGURU ‘kangaroo’ (Spencer2020).

In the Italian grammatical tradition, the same words, the adjective *invariabile* lit. ‘invariable’ and the noun *invariabilità* lit. ‘invariability’, are used for both cases. Traditional Italian grammars list nine parts of speech, of which five (articles, adjectives, nouns, pronouns and verbs) are said to be *variabili* ‘inflecting’ and four (adverbs, prepositions, conjunctions and interjections) *invariabili* ‘uninflecting’. Within the inflecting parts of speech, specific nouns and adjectives are *invariabili* ‘uninflectable’, i.e. appear in the same phonological form in any syntactic context, as the noun *città* ‘city, town’ in (??) and the adjective *blu* ‘blue’ in (??):

	a.	<i>l-a</i>	<i>città</i>	<i>vecchi-a</i>
(1)				
		DEF-F.SG	city(F)[SG] ‘the old town’	old- F.SG
	b.	<i>l-e</i>	<i>città</i>	<i>italian-e</i>
		DEF-F.PL	city(F)[PL] ‘the Italian cities’	Italian-F.PL

	a.	<i>l-a</i>	<i>penn-a</i>	<i>blu</i>
(2)				
		DEF-F.SG	pen(F)-SG 'the blue pen'	blue[F.SG]
	b.	<i>l-e</i>	<i>penn-e</i>	<i>blu</i>
		DEF-F.PL	pen(F)-PL 'the blue pens'	blue[F.PL]
	c.	<i>il</i>	<i>libr-o</i>	<i>blu</i>
		DEF.M.SG	book(M)-SG 'the blue book'	blue[M.SG]
	d.	<i>i</i>	<i>libr-i</i>	<i>blu</i>
		DEF.M.PL	book(M)-PL 'the blue books'	blue[M.PL]

In previous work (DresslerThornton1996, D'AchilleThornton2003, D'Achille2005a, D'AchilleThornton2008) we have investigated qualitative and quantitative properties of Italian uninflectable nouns. In this chapter, we illustrate uninflectability in Italian nouns, updating our previous findings on the basis of new data concerning usage in the first twenty years of the twenty-first century, and investigate adjectives with the same methodology used for nouns by D'AchilleThornton2003.¹

Spencer2020 also drew attention to *constructional non-inflectedness/ uninflectedness / uninflectability*, a property exhibited in “grammatical constructions involving lexemes which cannot bear inflections [...] in that context”, such as “the modifying non-head noun in an English noun-noun compound” (Spencer2020).² In this chapter, we will touch constructional uninflectability only briefly, in §??; a thorough investigation of this phenomenon in Italian must be left for further research.

¹Uninflectable Italian adjectives have been briefly discussed by Fedden2019 in the context of his investigation of “sporadic agreement”. To the best of our knowledge, they have not received specific attention in the context of Italian linguistics. Uninflectable verbs do not seem to exist in Italian.

²Spencer2020 uses “constructional non-inflectedness” in the title of section 4, but then discusses the possibility that we should make a terminological distinction between constructional uninflectedness (*sic*) and constructional uninflectability; this second term is considered more appropriate for cases in which a lexeme which normally inflects fails to do so in specific constructions (Spencer2020: 148–149). The Italian cases discussed in §??, then, are instances of constructional uninflectability.

The chapter is organized as follows: in §?? we present the inflectional classes of Italian nouns and adjectives, and quantitative data on types in each class from various available databases; in §?? we illustrate and discuss the factors that render Italian nouns and adjectives uninflectable, and their variation in diachrony, with a focus on the contemporary situation; in §?? we briefly discuss two cases of constructional uninflectability; §?? summarizes our conclusions.

2 Inflection and inflectional classes of Italian nouns and adjectives

Before delving into the investigation of Italian uninflectable nouns and adjectives, a brief sketch of Italian nominal inflection is in order.

Italian nouns inflect for **NUMBER**, which has two values: **SINGULAR** and **PLURAL**. They are usually assumed to have an inherent **GENDER** value, either **MASCULINE** or **FEMININE**. Nouns control agreement in Gender and Number on articles and adjectives (including possessives, demonstratives and some quantifiers) within NPs, and NPs control agreement in Gender and Number on predicates that agree (adjectives and past participles appearing in certain periphrastic verb forms, according to conditions described in Thornton2019: 78), on most third person pronouns, and on periphrastic relative pronouns.

Table ?? illustrates the paradigm of the Italian definite article, a reliable exponent of agreement feature values in most contexts, which will often appear in our examples below. In the singular, final vowels are deleted before vowel-initial words (this deletion is represented orthographically by an apostrophe); the distribution of the M.SG shapes *il* and *lo* and the M.PL shapes *i* and *gli* is mostly phonologically conditioned, but there are some lexically conditioned oddities (e.g., *gli dei* ‘the Gods’ vs. *i denti* ‘the teeth’; cfr. MaidenRobustelli2000: 62).³

Table 1: Forms and shapes of the Italian definite article

SINGULAR	<i>il, lo, l’</i>	<i>la, l’</i>
PLURAL	<i>i, gli</i>	<i>le</i>

Table ?? illustrates the inflectional classes of Italian nouns, according to the numbering proposed by D’AchilleThornton2003; class 7 is only attested in Old Italian, and is no longer used in contemporary Italian. Class 6 (highlighted by a bold frame) hosts uninflectable nouns.

³For a recent survey of the various proposals on the distribution of the shapes of the Italian M.SG definite article, see Repetti2020.

Table 2: Inflectional classes of Italian nouns

Nouns' endings		Class	Gender
SG	PL		
-o	-i	1	M with one exception
-a	-e	2	F
-e	-i	3	M, F
-a	-i	4	M with two exceptions
-o	-a	5	SG M, PL F
stressed V C -i ...	stressed V C	6	M, F
	...		
-o	-ora	7 (extinct)	SG M, PL F

The gender assignment system of Italian is mixed: most nouns denoting humans and animals that are economically important have semantic assignment (masculine is assigned to nouns denoting males, feminine to nouns denoting females), but there are a few epicene nouns denoting humans (e.g. *PERSONA* (F) ‘person’, *PERSONAGGIO* (M) ‘character (in a novel, play, etc)’ and many denoting animals (e.g. *GIRAFFA*(F) ‘giraffe’, *SERPENTE*(M) ‘snake’). Inanimate and abstract nouns are assigned a gender value on the basis of phonological or morphological criteria, or sometimes by semantic “crazy rules” (Enger2009; e.g., nouns denoting cars are feminine, cf. Thornton2001).

As shown in Table ??, classes 1, 2 and 4, and the extremely non-canonical and inquirate class 5 (on which cf. Thornton2019: 85–87) align with a gender value completely or with extremely few exceptions, while classes 3 and 6 host nouns of both genders. Table ?? shows the percentage of nouns of each gender in these classes.⁴

Table 3: Percentage of masculine and feminine nouns in classes 3 and 6

⁴Data in Table ?? are based on BDVDB (Thornton, IacobiniBurani1997), a database containing about 4500 nouns belonging to the Italian Basic Vocabulary (De Mauro1989).

	M	F
Class 3	50.7%	49.3%
Class 6	48.6%	51.4%

Table ?? illustrates the inflectional classes of Italian adjectives; for contemporary Italian we follow the classification proposed by **IacobiniThornton2016**, which contains classes 1-5; we have added classes 6 and 7 which are only attested in earlier stages of the language; the rightmost column of Table ?? draws attention to correspondences between inflectional classes of nouns and adjectives. As in Table ??, classes which host adjectives completely or partially uninflectable are highlighted with a bold frame. Classes 6 and 7 are only attested in Old Italian, and are no longer used in contemporary Italian.

Table 4: Inflectional classes of Italian adjectives

Adjectives' endings		Gender	Class	Corresponding class for nouns
SG	PL			
-o	-i	M	1	1
-a	-e	F		2
-e	-i	M=F	2	3
-a	-i	M	3	4
-a	-e	F		2
-e	-i	M	4	3
-a	-e	F		2
stressed V C -i ...	stressed V C	M=F	5	6
	-i			
-o	-a	M	6 (extinct)	5
-a	-e	F		2
-e	-i	M	7 (extinct)	3
-e	-e	F		6

In Tables 5 and 6 we present data on the percentage of lexemes in each class. The data are drawn from two different sources: BDVDB (Thornton, **IacobiniBurani1997**), a database containing information of various kinds on the about 7000 lexemes constituting the Italian Basic Vocabulary (De **Mauro1989**), and DeGNI (De Martino, **BraccoLaudanna2023**), a database containing information on gender and

inflectional class of the nouns contained in CoLFIS (BertinettoEtAl2005), a frequency dictionary of contemporary written Italian. BDVDB contains about 4500 nouns and about 1100 adjectives, i.e. those appearing among the top 5000 most frequent lexemes in LIF (Bortolini, TagliaviniZampolli1971, a frequency dictionary of written Italian based on a corpus of about 500,000 tokens, from texts published between 1947 and 1968), and those appearing among about 2500 lexemes not present in the top frequency ranks in LIF, but deemed to be highly available for speakers, since they denote objects and situations commonly encountered in everyday life (e.g., *forchetta* ‘fork’, *ingorgo* ‘traffic jam’, *carino* ‘cute’, *pigro* ‘lazy’); CoLFIS, on which DeGNI is based, contains over 20000 nouns, from a corpus of written Italian of about 3,8M tokens, drawn from press and literature published between 1992 and 1994.⁵ DeGNI contains only nouns; therefore, for adjectives we have only data from BDVDB (which contains 1129 adjectives), presented in Table ??.

Table 5: Percentage of nouns in each inflectional class (types)

Class	percentage of types in BDVDB N = 4583	percentage of types in CoLFIS / DeGNI N = 23619
1	37.7%	31%
2	34.4%	23%
3	20.8%	20%
4	1.3%	1%
5	0.3%	??
6	5.4%	21%

Table 6: Percentage of adjectives in each inflectional class (types)

⁵The data in Table ?? are not completely comparable with respect to class 5, because DeGNI does not consider it a separate class, and merges data on nouns belonging to this class with data on other inflectionally odd nouns, such as various types of compounds, in a category “Other”, which accounts for 4% of the nouns in the CoLFIS corpus on which DeGNI is based.

Class	percentage of types in BDVDB N = 1128
1	65.2%
2	31.7%
3	0.9%
4	0.01%
5	2.0%

Clearly, inflectional classes of Italian nouns and adjectives do not constitute a canonical system, as defined by Corbett (2009; see also Thornton2019: 79–87): membership in the various classes is not balanced, and uninflectable nouns and adjectives are few in BDVDB; however, in DeGNI, based on a corpus collected at the end of the twentieth century, class 6, containing uninflectable nouns, seems to be about as numerous as classes 2 and 3, which are inflecting. This seems to point to an increase of uninflectability of Italian nouns in recent times.

To investigate the evolution of inflectional class membership, we have created and manually annotated a small corpus, containing 500 tokens of nouns and 500 tokens of adjectives from Italian texts belonging to each one of eleven time periods, between 1211 and 2022, shown in Table ??.⁶ We have avoided as much as possible to establish the time periods mechanically, aligning them with centuries; whenever possible, we have chosen as initial and final dates for a given period years which are conventionally considered turning points in the linguistic history of Italian (such as Boccaccio’s death in 1375, the publication of Bembo’s *Prose della volgar lingua* in 1525, the publication of the first edition of the Italian dictionary by Accademia della Crusca in 1612, etc.). The twentieth century has been subdivided in three periods to have a better grasp of recent developments, and we have included texts from the twenty-first century up to 2022. For each time period, we have selected five prose texts belonging to different genres (such as novels and plays, scientific texts, essays, chronicles, ledgers, biographies and autobiographies, letters, legal texts, articles in newspapers and magazines) and have counted 100 tokens of nouns and 100 tokens of adjectives from a continuous portion of text, classifying each token in the inflectional class to which it belongs.

⁶The data about nouns in the periods between 1211 and 2000 have been collected in 2000 and analyzed in D’AchilleThornton2003; for the present chapter, we have extended the noun corpus to cover the period 2001–2022, and have collected and analyzed data on adjectives.

Table 7: Time periods used to partition the corpus analyzed

Time period	Dates
I	1211–1300
II	1301–1375
III	1376–1525
IV	1526–1612
V	1613–1691
VI	1692–1800
VII	1801–1900
VIII	1901–1945
IX	1946–1968
X	1969–2000
XI	2001–2022

3 Inflection and uninflectability of Italian nouns

3.1 3.1 Productivity of inflectional classes of Italian nouns

Previous qualitative and quantitative analyses (DresslerThornton1996, D’AchilleThornton2003) have shown that the various inflectional classes of Italian nouns have different degrees of productivity, defined in terms of Dressler’s (1997, 2003) criteria of inflectional productivity, according to which an inflectional class is productive at a given time if it gains new members through integration of loanwords and other types of neologisms or through class shift of lexemes changing their inflectional class in diachrony.⁷

Class 5 is completely unproductive (while it was productive at least up to the fourteenth century, according to Gardani2013), has been losing members to class 1 throughout the centuries (Santangelo1981), and is now inquorate (it contains around two dozens of nouns, many of which are overabundant in their plural cell, with a plural of class 1 alongside one of class 5, cf. Thornton2010–11).

Class 4 is losing members towards class 6, and is only fed by productive suffixation, most notably the formation of person-denoting nouns in *-ista* (such as *ter-rapiattista* ‘someone who believes that the earth is flat’ ↔ *terra piatta* ‘flat earth’).

Class 3 is losing members to class 6.

⁷See also Gardani2013 for a critical discussion of Dressler’s criteria.

Classes 1 and 2 are productive according to several criteria; they were productive even according to the strongest criterion in Dressler’s classification, integration of loanwords with unfitting phonological form, up to the nineteenth and even the early twentieth century (e.g., English *beefsteak*, “unfitting” because it ends in a consonant, was adapted as *bistecca*, attested since 1844, and is still inflected as a class 2 noun), but nowadays loanwords ending in consonants (i.e., with unfitting phonological form) are assigned to class 6. Nouns converted from verbs are assigned to class 1 (e.g., *scippo* ‘bag-snatching’ ↔ *scippare* ‘snatch [a bag]’) and some clipped nouns from deverbal suffixed action nouns are assigned to class 2 (e.g., *codifica* ↔ *codificazione* ‘coding’).

Class 6 is gaining members in several ways, mainly by hosting loanwords ending in a consonant or in /i/; even indigenous affixation feeds it, e.g. with quality nouns in *-ità* such as *padanità* ‘quality of belonging to the Po valley territory [Val Padana]’.

Table ?? shows the increase of membership in class 6 of both tokens and types in our corpus, from the thirteenth to the end of the twentieth century; this increase has been shown to be statistically significant (D’AchilleThornton2003: 219).

Table 8: Percentage of types and tokens of nouns in class 6 in the corpus

	13th century	end of 20th century
tokens	2.4%	8.6%
types	2.7%	9.5%

D’AchilleThornton2003 concluded that the main system-defining property of Italian noun inflection is the alignment of back vowels (/a/ and /o/) with SINGULAR and of front vowels (/e/ and /i/) with PLURAL, and the alignment of the singular ending /a/ with FEMININE and the singular ending /o/ with MASCULINE; nouns not displaying such an alignment in their paradigm, such as class 3 nouns, that end in the front vowel /e/ in the singular and can belong to either gender (cf. Table ??), so that some plurals in /i/ are feminine, show an increasing tendency to become uninflectable, as do masculine nouns ending in /a/ (class 4) and feminine nouns ending in /o/ (investigated by D’AchilleThornton2008).

3.2 Properties of uninflectable Italian nouns

In this section we illustrate the properties that influence the uninflectability of nouns in Italian. These properties are mainly phonological, but the relation be-

tween a noun's gender and its ending in the singular also plays a role, as do orthographic and stratal properties.

Table ?? shows the progressive appearance of nouns of various kinds in our corpus. A $\square$ in a cell in Table ?? means that nouns of the kind listed in the row are attested in the time period corresponding to the column. A continuous shading means that nouns of the kind listed in the row are unattested in the time periods corresponding to the columns, and are only attested later; the lighter shading in the row corresponding to nouns ending in a consonant is due to the fact that there is a doubtful occurrence of a noun ending in a consonant in period I (*dies* 'day', which, however, is a Latin word inserted in a vernacular context, a kind of code-switch), but then no more tokens till period VIII, that is till the twentieth century. An empty white cell means that nouns of the kind listed in the row are not attested in our corpus in the time period corresponding to the column, but are attested in the corpus in earlier periods.

INSERT TABLE 9 ABOUT HERE

3.2.1 3.2.1 Phonological properties

The main property that makes a noun uninflectable for phonological reasons in Italian is ending in a stressed vowel.⁸ Nouns ending in stressed vowels and treated as uninflectable are attested in all periods in our corpus (cf. Table ??), and have existed throughout the history of the Italian language. They have come about at first from Latin nouns, in various ways: some inherited nouns have come to end in a stressed vowel because a final consonant or consonant cluster has been lost (??); a sizeable number of nouns end in a stressed vowel after loss by apocope of a final unstressed syllable (??); another source of final stressed vowels is the deletion of a final /e/, which was etymological but was reanalyzed as epithetic and consequently deleted (??).

(3)

a. Lat. REX > *re* 'king'

b. Lat. CIVITATE(M) > *cittate/ cittade* > *città* 'city'

Lat. VIRTUTE(M) > *virtute / virtude* > *virtù* 'virtue'

⁸In the standard Italian orthography, final stressed vowels are marked by an accent in polysyllabic native words and orthographically integrated loanwords, and in some but not all monosyllables, according to various normative rules which we will not illustrate here because they are orthogonal to the issue that concerns us. In the text we follow standard spelling, and provide phonological transcriptions when necessary.

c. Lat. DIE(M) > *die* > *dì* ‘day’

Later the type has grown through loanwords, such as the ones in (??), encountered in our corpus:

(4)

falpalà ‘flounce’, *caffè* ‘coffee’

There has never been any attempt at integrating nouns ending in a stressed vowel in one of the inflecting classes. These are the nouns most truly uninflectable for phonological reasons.

A second group of nouns whose uninflectability correlates with a phonological property is represented by nouns ending in /i/.⁹ These are attested in our corpus since period V, roughly corresponding to the seventeenth century. This type at first included only feminine nouns

of Greek origin, such as the ones in (??):

(5)

ipotesi ‘hypothesis’, *estasi* ‘ecstasy’, *crisi* ‘crisis’...

Later, the type has grown through loanwords, most often

assigned masculine gender, such as the ones in (??) from our corpus:

(6)

baby /bɛbi/ ‘kid’¹⁰, *floppy* ‘floppy disk’...

In masculine nouns frequently used in the plural, the final /i/ can be perceived as a M.PL ending of class 1, with consequent backformation of a singular in /o/; this happened in the case of *cachi* < Jap. *kaki* ‘Diospyros kaki’.¹¹ Table ?? shows the frequency of the two options in the *ItTenTen20* corpus (a corpus which is part of the *TenTen* corpus family, and contains 12.4 billion tokens from texts downloaded from the Web in the years 2019-2020) in contexts in which the noun is used in the singular preceded by the indefinite article:

⁹We do not address uninflectable nouns ending in unstressed /u/, which are very few and quite recent loanwords such as *guru*, *haiku*, *sudoku*, *tofu*.

¹⁰In the context in which it appears in our corpus, *i baby* refers to football players in a junior team, and could be glossed ‘the colts’.

¹¹In theory, a masculine plural form in *-i* could lead to backformation of a class 3 singular in *-e* or a class 4 singular in *-a*: however, such cases are not attested in the thorough investigation of backformation in Italian by D’Achille2005b.

Table 9: Inflectional forms of singular *cachi* / *kaki* in *ItTenTen20*

uninflectable (class 6)	tokens	inflected in class 1	tokens
<i>un cachi</i>	106	<i>un caco</i>	203
<i>un kaki</i>	30	<i>un kako</i>	7

Inflection in class 1 through backformation clearly correlates with orthographic integration: in the standard spelling of contemporary Italian the letter <k> is only used in loanwords, and the data show that the most common option is inflection in class 1 for the orthographically integrated forms (with a ratio of 1.9 : 1 in favour of class 1), but uninflectability for the forms spelled with <k> (with a ratio of 4.3 : 1 in favour of class 6).

The last phonological property that makes a noun uninflectable in contemporary Italian is its ending in a consonant. In our corpus, this type appears only in the twentieth century, and is attested in all time periods since the VIII, with nouns such as the ones in (??), which are mostly loanwords (but a few, such as the brand name *domopak* ‘plastic food wrap’, are autochthonous formations):

(7)

lastex, rayon, sport, tweed, élités, toilettes, claqué, refrain, soubrette, round, goal, supporter, assist, vermouthe, Nord, Internet, computer, hardware, software, hard disk, mouse, monitor, modem, scanner, mail, domopak, hashish, babysitter, performance, governance, input

Previously, loan nouns ending in a consonant were adapted, appending a final unstressed vowel (/a/ for feminine nouns, /o/ for masculine ones) after gender assignment; only rarely, a final /e/ was supplied, either epenthetically or because the loanword was analyzed as containing a suffix ending in -e. These nouns were then inflected in the class appropriate to their gender and ending. This is shown by the data in (??), with examples of Arabic (from Mancini1992) and English loanwords integrated before the twentieth century.

(8)

a. Arabic loanwords

nouns denoting human males integrated in class 1:

sulṭān > *sultano* ‘sultan’